Midterm Review

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High-Level Review

The midterm will mainly cover linear optimization

- Formulate a problem into linear optimization
- Simplex method
- Duality theories
- Sensitivity analysis

Basics about Linear Optimization

Know the standard form of LP and be able to transform any LP to standard form:

$$\begin{aligned} \text{minimize}_{\mathbf{x}} \quad & \mathbf{c}^T\mathbf{x} \\ \text{subject to} \quad & A\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq 0 \end{aligned}$$

Techniques to transform non-standard form to standard form

- Maximization problem
- Inequality constraints: add slack variables
- Free variables: write as the difference of two non-negative variables

Transform to LP

We discussed several types of scenarios where a seemingly non-LP can be transformed to an LP

- Maximin/minimax objective
- Absolute values (in either objective function or constraints)
- You should know why those tricks work and be able to apply them appropriately

Geometry of LP

You should be able to solve a small LP using its geometric representation (when it is possible)

Understand the concepts of

- Extreme points
- Basic solutions
- Basic feasible solutions (BFS)
- Basis/non-basis
- Basic/non-basic variables

For small LPs, can identify the basic solutions and basic feasible solutions.

Facts about BFS

Theorem

For the standard LP polyhedron $\{\mathbf{x} : A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq 0\}$*, the followings are equivalent:*

1. $\mathbf{x}$ *is an extreme point*
2. $\mathbf{x}$ *is a basic feasible solution*

Theorem

For any linear optimization problem where the constraint matrix has full rank:

- *If there is a feasible solution, then there is a basic feasible solution;*
- *If there is an optimal solution, then there is an optimal solution that is a basic feasible solution.*

Facts about LPs

Corollary

In order to find an optimal solution, we only need to look among basic feasible solutions (no more than $C(n, m)$ of them).

Corollary

If an LP with m constraints (in the standard form) has an optimal solution, then there must be an optimal solution such that there is no more than m positive entries.

Simplex Method

- Understand how simplex method works
- Can use simplex tableau to solve small LPs
- Can read and interpret simplex tableau
- Can use two-phase method

The High-Level Idea of Simplex Method

The simplex method searches among BFS. It proceeds from one BFS to a neighboring one, in such a way as to continuously improve the value of the objective function until it reaches optimality.

Definition

Two basic solutions are neighboring if they differ by exactly one basic index

Simplex Method

When at a BFS, the simplex method considers its neighbors by choosing one entering basis.

- When the incoming index is chosen, the (jth feasible) direction is also set:

$$\mathbf{d} = [-A_B^{-1}A_j; 0; ..; 1; ..; 0]$$

- The change to the objective value when going in jth feasible direction (in a unit step size) is given by the reduced cost $\bar{c}_j$

$$\bar{c}_j = c_j - \mathbf{c}_B^T A_B^{-1} A_j$$

- When all the reduced costs are non-negative, the current BFS is optimal

Stepsize

If there is a negative reduced cost, then one can go along that direction and the objective value will decrease.

Then we need to choose the step size

- We want to go as far as possible without violating any $\mathbf{x} \geq 0$ constraint
- If $\mathbf{d} \geq 0$, then there is no limit on how much one can go, and the problem is unbounded
- Otherwise one can only go

$$\theta^* = \min_{\{i \in B | d_i < 0\}} \left(-\frac{x_i}{d_i} \right)$$

One Iteration of the Simplex Method

We start from a BFS $\mathbf{x}$

1. We first compute the reduced costs $\bar{\mathbf{c}}$
   - If no reduced costs is negative, we are already optimal
   - Otherwise choose some j such that $\bar{c}_j < 0$
2. Compute the jth basic direction

$$\mathbf{d} = [-A_B^{-1}A_j; 0; ...; 1; ...; 0]$$

   - If $\mathbf{d} \geq 0$, then the problem is unbounded, i.e., the optimal value is $-\infty$.
   - Otherwise, compute θ^*
3. Define $\mathbf{y} = \mathbf{x} + \theta^*\mathbf{d}$. And repeat these procedures

Degeneracy

Degenerate case happens when there exists basic variable that equals 0

- To ensure there is no cycling, we could use the smallest index rule when choosing the incoming and outgoing basis
- If the smallest index rule is used, one needs not to worry about the degenerate cases in the simplex method

Simplex Tableau

Bottom line: be able to solve LP using simplex tableau.

Understand each part of the simplex tableau:

$\mathbf{c}^T - \mathbf{c}_B^T A_B^{-1} A$	$-\mathbf{c}_B^T A_B{}^{-1} \mathbf{b}$
$A_B^{-1} A$	$A_B^{-1} \mathbf{b}$

- The upper left corner is the reduced cost vector
- The upper right corner is the negative of the objective value
- The bottom part is the transformed constraints and the corresponding basic variables
- One can find $A_B^{-1} A$ in the tableau. In particular, one can often find A_B^{-1}. When one started from the canonical form, then A_B^{-1} is the columns where the initial basis was (usually the last few columns if those correspond to slack variables).

Canonical Form

During the simplex tableau operations, it always maintains a canonical form (from the start to the end)

- Contains an identity matrix
- The reduced costs for the basic part is zero

In a canonical form, it is easy to tell the current BFS.

Pivoting Procedures

From a canonical form, each iteration in the simple tableau operation consists of

1. Determine optimality: look at the top row (the reduced costs)
   - If all non-negative, then the current solution is optimal
   - Otherwise, choose the smallest index j such that $\bar{c}_j < 0$ to be the incoming basis
2. Perform the minimal ratio test (between the current basic variables and the incoming column - positive entry only).
3. Choose the row that attains the smallest ratio to be the pivot row, and determine the outgoing basis (smallest index rule)
4. Update the tableau
   - Divide each entry in the pivot row by the pivot element
   - Add proper multiples of the new pivot row to each other rows, such that all other elements in the pivot column become zeros (including the top row)

Simplex Tableau

- If there is no positive entries in the incoming column, then the problem is unbounded
- The update of the tableau should always include the top row and also the right hand side **b**

Two-Phase Method

The goal of the two-phase method is to find an initial BFS for the original problem.

In the two-phase simplex method, we first solve an auxiliary problem ($\mathbf{e}$ means an all-one vector, $\mathbf{b} \geq 0$).

$$\begin{aligned} \text{minimize}_{\mathbf{x},\mathbf{y}} \quad & \mathbf{e}^T\mathbf{y} \\ \text{subject to} \quad & A\mathbf{x} + \mathbf{y} = \mathbf{b} \\ & \mathbf{x}, \mathbf{y} \geq 0 \end{aligned}$$

- Solving this auxiliary problem will give one a BFS for the original problem (or tell that the original problem is infeasible)

Simplex Tableau for the Two-Phase Method

Construct the tableau for phase I problem. One needs to compute the reduced costs for the non-basic part (non-auxiliary variable), which is given by the negative of the sum of each columns

After solving the phase I problem, obtain the final tableau. Drop the columns of the auxiliary variables. Keep the other part of the constraints and the right-hand side vector. Recompute the reduced costs using the original objective coefficients (only need to compute the non-basic part):

$$\mathbf{c}_N - \mathbf{c}_B^T A_B^{-1} A_N$$

If the optimal solution to the phase I problem contains auxiliary variables (must be degenerate), then one needs to replace the auxiliary variable by any non-auxiliary variable to form a basis, which will give an initial BFS for the original problem

Duality Theory

- Be able to construct the dual for any linear program
- Know the duality theorems and can use them in different situations
- Be able to write down the complementarity conditions and use them
- Sensitivity analysis

Construct the Dual

Primal	minimize	maximize	Dual
constraints	$\geq b_i$ $\leq b_i$ $= b_i$	≥ 0 ≤ 0 free	variables
variables	≥ 0 ≤ 0 free	$\leq c_j$ $\geq c_j$ $= c_j$	constraints

Basic rules:

1. Each constraint of the primal problem corresponds to a dual variable
2. Each variable of the primal problem corresponds to a dual constraint
3. Equality constraints always correspond to free variables, vice versa
4. For the signs of the inequalities, *usual (unusual)* constraints always correspond to *usual (unusual)* constraints

Weak Duality Theorem

Theorem (Weak Duality Theorem)

If $\mathbf{x}$ *is feasible to the primal problem and* $\mathbf{y}$ *is feasible to the dual problem, then*

$$\mathbf{b}^T\mathbf{y} \leq \mathbf{c}^T\mathbf{x}$$

- Primal (dual) feasible solution can be used to construct bounds for the objective value of the dual (primal) problem
- If one is unbounded, then the other one must be infeasible
- Let $\mathbf{x}$ and $\mathbf{y}$ be feasible solutions to the primal and dual problems respectively. If $\mathbf{c}^T\mathbf{x} = \mathbf{b}^T\mathbf{y}$, then $\mathbf{x}$ and $\mathbf{y}$ are optimal solutions to the primal and dual, respectively.

Strong Duality Theorem

Theorem (Strong Duality Theorem)

If a linear program has an optimal solution, so does its dual, and the optimal value of the primal and dual are equal

All possible states of a pair of primal/dual linear programs

	Finite Optimum	Unbounded	Infeasible
Finite Optimum	$\surd$		
Unbounded			$\surd$
Infeasible		$\surd$	$\surd$

Optimality Conditions

Optimality conditions for LP: $\mathbf{x}$, $\mathbf{y}$ are optimal to the primal and dual problems respectively if and only if:

- $\mathbf{x}$ is primal feasible
- $\mathbf{y}$ is dual feasible
- $\mathbf{c}^T\mathbf{x} = \mathbf{b}^T\mathbf{y}$

Complementarity Conditions

- Be able to write the whole set of complementarity conditions
- Be able to use them to solve the primal/dual problem given the solution of the other (when it is possible)
- Be able to conclude meaningful conclusions using them

Complementarity Conditions

Consider the primal-dual pair:

Primal			**Dual**		
minimize	$\mathbf{c}^T\mathbf{x}$		maximize	$\mathbf{b}^T\mathbf{y}$	
subject to	$\mathbf{a}_i^T\mathbf{x} \geq b_i,$	$i \in M_1,$	subject to	$y_i \geq 0,$	$i \in M_1$
	$\mathbf{a}_i^T\mathbf{x} \leq b_i,$	$i \in M_2,$		$y_i \leq 0,$	$i \in M_2$
	$\mathbf{a}_i^T\mathbf{x} = b_i,$	$i \in M_3,$		y_i free,	$i \in M_3$
	$x_j \geq 0,$	$j \in N_1,$		$A_j^T\mathbf{y} \leq c_j,$	$j \in N_1$
	$x_j \leq 0,$	$j \in N_2,$		$A_j^T\mathbf{y} \geq c_j,$	$j \in N_2$
	x_j free,	$j \in N_3,$		$A_j^T\mathbf{y} = c_j,$	$j \in N_3$

Theorem

Let $\mathbf{x}$ *and* $\mathbf{y}$ *be feasible solutions to the primal and dual problems respectively. Then* $\mathbf{x}$ *and* $\mathbf{y}$ *are optimal if and only if*

$$y_i \cdot (\mathbf{a}_i^T\mathbf{x} - b_i) = 0, \quad \forall i; \qquad x_j \cdot (A_j^T\mathbf{y} - c_j) = 0, \quad \forall j$$

Optimality Conditions in Another Form

Now with the complementarity conditions, we can write an equivalent set of optimality conditions for LP:

1. $\mathbf{x}$ is primal feasible
2. $\mathbf{y}$ is dual feasible
3. All complementarity conditions are satisfied

Interpret the Dual Problem

Understand the examples we discussed in class and can interpret similar problems with proper hints:

- Production planning problem
- Multi-firm alliance problem
- Maximum flow problem
- Alternative systems

Sensitivity Analysis

Let

$$\begin{aligned} V = \min_{\mathbf{x}} \quad & \mathbf{c}^T\mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq 0 \end{aligned}$$

Then $\nabla V(\mathbf{b}) = \mathbf{y}^*$ if the dual problem has a unique optimal solution $\mathbf{y}^*$ and $\nabla V(\mathbf{c}) = \mathbf{x}^*$ if the primal problem has a unique optimal solution $\mathbf{x}^*$.

- If we change b_i by a small amount Δb_i, then the change of objective value will be $\Delta b_i y_i^*$
- If we change c_i by a small amount Δc_i, then the change of objective value will be $\Delta c_i x_i^*$
- The results hold for general LPs (not only standard form)

Sensitivity Analysis

The above analysis only holds for small changes

- Small change means that the optimal basis does not change
- If the optimal basis changes, then the change of optimal value and solution may be quite different, usually requires much more efforts

It is useful to know what range of change is a "small change" so that the local sensitivity analysis will work

Sensitivity Analysis

Case 1: $\mathbf{b}$ changes to $\tilde{\mathbf{b}} = \mathbf{b} + \Delta\mathbf{b}$

- The reduced costs do not change
- Therefore, in order to make sure that the optimal basis is unchanged, we only need to guarantee that

$$\tilde{\mathbf{x}}_B = A_B^{-1}(\mathbf{b} + \Delta\mathbf{b}) = \mathbf{x}^* + A_B^{-1}\Delta\mathbf{b} \geq 0$$

- If we only change one component of $\mathbf{b}$ by λ, then we can solve the range of λ from a set of linear inequalities

Sensitivity Analysis Continued

Case 2: $\mathbf{c}$ changes to $\tilde{\mathbf{c}} = \mathbf{c} + \Delta\mathbf{c}$

- The basic solution (corresponding to the same basis) does not change
- Therefore, in order to make sure that the current basis is still optimal, we only need to guarantee that the reduced costs are still non-negative, i.e.,

$$\tilde{\mathbf{c}}_N - A_N^T (A_B^{-1})^T \tilde{\mathbf{c}}_B \geq 0$$

  We only need to compute the reduced costs for the non-basic part, since the reduced costs for the basic part will still be 0
- If we only change one component of $\mathbf{c}$ by λ, then we can solve the range of λ from a set of linear inequalities

What if the Change is Outside the Range

If the change of $\mathbf{c}$ is so much that the reduced cost of the current solution contains negative number, then

- We can continue with the simplex tableau until it reaches optimal solution.

If the change of $\mathbf{b}$ is so much that the solution corresponding to the original optimal basis B is no longer feasible

- Then we may need to solve the problem from the start
- It can be viewed as that the objective coefficients of the dual problem changed. Then one can use the method that deals with changes in objective coefficients

Changes to A

If the change is for a number in a non-basic column, say A_j, then the original optimal solution is still feasible, the only change is to the reduced cost of jth variable.

- Recompute $\bar{c}_j$. If it is still non-negative, then the original optimal solution stays optimal. Otherwise, update the tableau for the jth column as well as the reduced cost and continue from there.

If the change is for a number in a basic column, then nearly all the numbers in the tableau will change. In general, there is not a simple way to deal with it.

Other Changes

Adding a variable (the rest are kept the same):

- The original BFS is still BFS, the reduced cost is unchanged.
- Only need to check the reduced cost corresponding to the new variable. If it is non-negative, then the original optimal solution is still optimal; otherwise continue the simplex method from there.

Adding a constraint:

- If the original optimal solution satisfies the constraint, then it is still optimal.
- If not, then the best way to deal with it is to think it as adding a dual variable, then use the simplex tableau for the dual problem to continue calculations.

Complexity Theory

Complexity of an algorithm

- The number of arithmetic operations the algorithm takes to solve a problem.
- It only concerns about the worst-case instance. Average case may perform better
- Polynomial vs exponential algorithm

Definition (Complexity of a Problem)

For a problem, if there exists a polynomial time algorithm, then we say it is in P.

Complexity of LP:

- No configuration of simplex method is known to be a polynomial algorithm
- Interior point method is a polynomial time algorithm. Therefore, LP is in P.

Interior Point Method

Only need to know the high level concept.

The idea comes from the optimality conditions:

1. (Primal Feasible) $A\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq 0$
2. (Dual Feasible) $A^T\mathbf{y} \leq \mathbf{c}$
3. (Complementarity) $x_i \cdot s_i = x_i \cdot (c_i - \mathbf{a}_i^T\mathbf{y}) = 0$ for each i

- Simplex method always keeps 1 and 3, and looks for solutions that satisfy 2;
- Dual simplex method always keeps 2 and 3, and looks for solutions that satisfy 1;
- Interior point method always keeps 1 and 2, and looks for solutions that satisfy 3.

Interior Point Method

Properties of interior point method:

- It always searches in the interior of the feasible region.
- The optimal solution output by the interior point solution is not necessarily a BFS (however, if there is only a unique optimal solution, then it still must be a BFS).
- If the optimal solution is not unique, then the interior point method will always output the one with the highest rank (the solution with the maximum number of non-zeros).